Biodiversity and Conservation

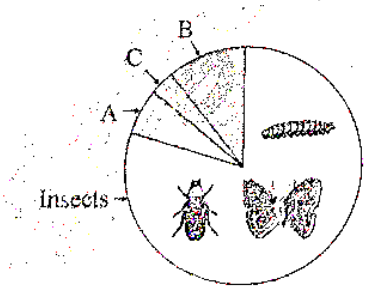
Introduction and Biodiversity

1. Species diversity
1) Is diversity of a species at genetic level
2) Is the diversity of a species within a region
3) Plays a role in process of speciation
4) Enables a population to adapt to its environment

2. What percentage of global biodiversity is shared by India?
1) 8.1%
2) 2.4%
3) 12%
4) 10%

3. Select the incorrect statement.
1) More than 70% of all species are animals.
2) Amphibian species are more diverse in Eastern Ghats than in Western Ghats.
3) Plant comprises not more than 22% of the total species diversity of the world.
4) Number of fungal species in the world is more than combined total of fishes, amphibians, reptiles and mammals.

4. Identify A, B and C in the following representation of global biodiversity showing proportionate number of species of major taxa of invertebrates.



	A	B	C
1)	Molluscs	Crustaceans	Other animal group
2)	Molluscs	Other animal groups	Crustaceans
3)	Crustaceans	Molluscs	Other animal group
4)	Other animal groups	Crustacéans	Molluscs

5. Given below are two statements.
Statement I: If an alien from a distant galaxy were to visit our planet Earth, the first thing that would amaze and baffle him would most probably be the enormous diversity of life that he would encounter.
Statement II: Even for humans, the rich variety of living organisms with which they share this planet never ceases to astonish and fascinate us.
In the light of the above statements, choose the most appropriate answer from the options given below:
1) Both statements I and II are incorrect.
2) Statement I is correct, but statement II is incorrect.
3) Statement I is incorrect, but statement II is correct.
4) Both statements I and II are correct.

6. India has 1000 varieties of mango with different size colour and taste due to

1) Genetic diversity
2) Species diversity
3) Omega diversity
4) Alpha diversity

7. Biogeographic region in India. having maximum endemism for amphibians is
1) Western Ghats
2) Himalaya
3) Deccan Peninsula
4) Gangetic Plain

8. How do ecologist estimate the total number of species present in the world?
1) By counting them all
2) By counting the number of any one species and applying this to various genus
3) By making statistical comparison of the temperate tropical species richness of an exhaustively studied group of insects and extrapolating this rate to other group of plants and animals
4) By calculating biomass of all the species

9. Species-area relationship was given by
1) Alexander von Humboldt
2) David Ehrlich
3) Paul Ehrlich
4) Robert Costanza

10. Reserpine which is an active chemical drug is obtained from
1) Atropa
2) Papaver
3) Datura
4) Rauwolfia

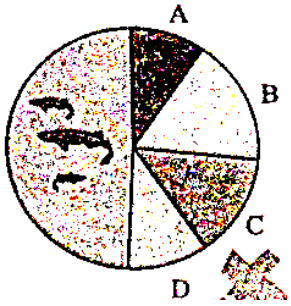
11. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Biodiversity exists at all the levels of biological organisations ranging from macromolecules within cells to biomes.
Reason (R): The term 'biodiversity' describes the combined diversity at all the levels of biological organisations.
In the light of the above statements, choose the most appropriate answer from the options given below:
1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
2) (A) is correct but (R) is not correct.
3) (A) is not correct but (R) is correct.
4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

12. In the formula log S = log C+ Z log A, ecologist have discovered that the value of Z usually lies in the range of
1) 0.1 to 0.5
2) 5 to 10
3) 0.1 to 0.2
4) 1 to 5

13. Tropical regions have greater biological diversity than temperate region
A. Because of constant and favorable environment throughout the year in tropical regions.
B. Due to low temperature and less humidity in tropics.
1) Only (A) is correct

2) Only (B) is correct
3) Both (A) and (B) are correct
4) Both (A) and (B) are incorrect

14. Identify A, B, C and D in the given pie chart representing global biodiversity of vertebrates.



	A	B	C	D
1)	Birds	Mammals	Reptiles	Amphibians
2)	Amphibians	Birds	Reptiles	Mammals
3)	Mammals	Birds	Reptiles	Amphibians
4)	Reptiles	Mammals	Amphibians	Birds

15. Species diversity _____ as we move away from the equator towards the poles.
Fill in the blank.
1) Increases
2) Decreases
3) First increases then decreases
4) First decreases then increases

16. The relation between species richness and area for a wide variety of taxa (angiosperm plants, birds, bats, freshwater fishes) turns out be a
1) Rectangular parabola
2) Rectangular hyperbola
3) Straight line/linear
4) Zig zag

17. The IUCN Red List (2004) documents the extinction of total how many species?
1) 87 species
2) 784 species
3) 27 species
4) 358 species

18. Rivet popper hypothesis was given by
1) Alexander von Humboldt
2) Edward Wilson
3) Paul Ehrlich
4) David Tilman

19. Select the mismatched pair:
1) Alexander von Humboldt: Species-area relationship
2) David Tilman: Outdoor plots
3) Robert May: Global species diversity
4) Paul Ehrlich: The term biodiversity

20. Given below are two statements.
Statement I: The biological wealth of our planet has been declining rapidly and the accusing finger is clearly pointing to human activities.
Statement II: It is doubtful if any new species are being added (through speciation) into the earth's treasury of species, there is no doubt about their continuing losses.

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Both statements I and II are incorrect.
2) Statement I is correct, but statement II is incorrect.
3) Statement I is incorrect, but statement II is correct.
4) Both statements I and II are correct.

21. Which of the following may not be a reason for extinction of biodiversity?

1) Habitat loss and fragmentation
2) Alien species invasion
3) Keeping animals in zoological parks
4) Overexploitation of natural resources

22. "In rivet popper hypothesis" of Paul Ehrlich, the rivets on the wings represent ____ in ecosystem.

1) Key species
2) Biome
3) Extinct species
4) Habitat

23. All the given characteristics are related to susceptibility of species to extinction except

1) Small population size
2) Large body size
3) Fixed habitat and migratory routes
4) High reproductive rate

24. The IUCN Red List (2004) documents the extinction of 784 species including

1) 338 vertebrates, 359 invertebrates and 87 plants
2) 359 vertebrates, 338 invertebrates and 87 plants
3) 338 vertebrates, 259 invertebrates and 187 plants
4) 359 vertebrates, 238 invertebrates and 187 plants

25. Extinction of wasp population will also lead to extinction of fig species. This statement indicates towards which of the following cause of biodiversity loss?

1) Overexploitation
2) Coextinction
3) Alien species invasion
4) Habitat loss

26. What does 'stability' mean in context to a biological community?

1) A stable community should not show too much variation in productivity from year to year.
2) A stable community should not show too much variation in productivity from year to year.
3) A stable community must be resistant to invasion by alien species.
4) All ofthe above

27. When the last member of a particular species dies, the species is said to be _____.

1) Critically endangered
2) Endangered
3) Diversified
4) Extinct

28. Red data book provides data on

1) Red flowered plants
2) Red coloured fishes
3) Red eyed birds
4) Endangered plants and animals

29. The organization which publishes the Red list of species 15

1) BSI
2) ICSI
3) WPSI
4) IUCN

30. IUCN Red list (2004) documents the extinction of A species in the last B years.

	A	B
1)	500	300
2)	784	500
3)	654	500
4)	984	200

Biodiversity Conservation

31. All of the following are ex situ biodiversity conservation strategies, except
1) Seed Banks
2) Zoological Parks
3) Botanical Gardens
4) National Parks

32. How many of the following are broadly utilitarian reasons for conserving biodiversity?
(i) Oxygen
(ii) Pollination
(iii) Flood and erosion control
(iv) Fibres and wood
(v) Drugs
(vi) Aesthetic pleasure
1) Two
2) Three
3) Four
4) Five

33. Amazon rain forest are considered as lungs of the planet because they contribute _____ of the total oxygen in the earth's atmosphere.
1) 5%
2) 10%
3) 20%
4) 30%

34. "Hotspots of biodiversity" show
1) High degree of endemism
2) High levels of species richness
3) Accelerated habitat loss
4) All of the above

35. How many among the following are involved in the conservation of endangered species at the place out of their natural habitat?

National parks, sacred groves, botanical garden, wildlife safari parks, biosphere reserves, seed bank, zoological parks, wildlife sanctuaries

1) Five
2) Four
3) Six
4) Three

36. Given below are two statements.

Statement I: Nobody knows how many more medicinally useful plants are there in the tropical rain forests, waiting to be explored.

Statement II: With increasing resources put into 'bioprospecting' nations endowed with rich biodiversity can expect to reap enormous benefits.

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Both statements I and II are incorrect.
2) Statement I is correct, but statement II is incorrect.
3) Statement I is incorrect, but statement II is correct.
4) Both statements I and II are correct.

37. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): In many cultures of India, tracts of forest were set aside, and all the trees and wildlife within were venerated and given total protection.

Reason (R): India has a history of religious and cultural traditions that emphasised protection of nature.

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
2) (A) is correct but (R) is not correct.
3) (A) is not correct but (R) is correct.
4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

38. Read the following statements.

A. Norman Myers developed the concept of hotspots to designate priority areas for ex situ conservation.
B. The hotspots are the richest and the most threatened reservoirs of plants' and animals' life on the earth.
C. The key criteria for determining a hotspot are low endemism and high level of species richness.

The correct statement is/are

1) (A) and (B)
2) (A) and (C)
3) Only (B)
4) (A), (B) and (C)

39. The number of National Parks (A), Wildlife Sanctuaries (B), and Biosphere Reserves (C), of India are

	A	**B**	**C**
1)	90	448	14
2)	20	448	20
3)	58	412	14
4)	78	212	18

40. The World Summit on sustainable development was held in A at B

Fill the blanks by selecting the correct option for A and B, respectively.

	A	**B**
1)	2002	Johannesburg, South Africa
2)	2002	Rio de Janeiro, Brazil
3)	1992	Johannesburg, South Africa
4)	1992	Rio de Janeiro, Brazil

41. Earth Summit, 1992 was held at

1) Rio de Janeiro, Brazil
2) Doha, Qatar
3) Kyoto, Japan
4) Johannesburg, South Africa

42. Read the following statements and choose the correct option.

I. The reasons for conserving biodiversity are categorized into narrowly utilitarian, broadly utilitarian, and ethical reasons.

II. Besides the direct benefits (food, fibre, firewood, pharmaceuticals, etc.), there are many indirect benefits we receive through ecosystem services such as pollination, pest control, climate moderation, and flood control.

III. Recently, 25 'biodiversity hotspots' in the world have been proposed for intensive conservation efforts. Of these, three (Western Ghats-Sri Lanka, Himalaya, and Indo-Burma) cover India's rich biodiversity regions.

IV. Biodiversity refers to the sum total of diversity that exists at all levels of biological organisation.

Correct statement(s) is/are

1) All of the above
2) I, II and III
3) II, III and IV
4) I. II and IV

43. How many hotspots of biodiversity in the world have been identified?

1) 25
2) 24
3) 40
4) 34

44. When plants and animals are protected in their natural babitats it is called

1) In situ conservation
2) Ex situ conservation
3) Cryopreservation
4) Greenhouses

45. Given below are two statements.

Statement I: The ethical argument for conserving biodiversity relates to what we owe to millions of plant, animal, and microbe species with whom we share this planet.

Statement II: Philosophically or spiritually, we need to realise that every species has an intrinsic value, even if it may not be of current or any economic value to us

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Both statements I and II are incorrect.
2) Statement I is correct, but statement Il is incorrect.
3) Statement I is incorrect, but statement Il is correct.
4) Both statements I and II are correct.

46. Sacred grooves are especially useful in

1) Generating environmental awareness
2) Preventing soil erosion
3) Year-round flow of water in rivers
4) Conserving rare and threatened species

47. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The narrowly utilitarian arguments for conserving biodiversity are obvious.

Reason (R): The broadly utilitarian argument says that biodiversity plays a major role in many ecosystem services that nature provides.

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
2) (A) is correct but (R) is not correct.
3) (A) is not correct but (R) is correct.
4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

48. More than _____ % of the drugs currently sold in the market worldwide are derived from plants.

1) 70
2) 25
3) 90
4) 55

49. All are examples of broadly utilitarian, except

1) Pollination
2) Oxygen
3) Aesthetic pleasure
4) Firewood

50. Ex situ conservation includes

1) Cryopreservation of gametes
2) In vitro fertilization
3) Tissue culture techniques
4) All of these

51. Read the following statements and choose the correct option.

I. The hotspots are also regions of accelerated habitat loss.

II. In recent years, in situ conservation has advanced beyond keeping threatened species in enclosures.

III. Biodiversity knows no political boundaries and its conservation is therefore a collective responsibility of all nations,

IV. Initially 34 biodiversity hotspots were identified but subsequently nine more have been added to the list.

Correct statement(s) is/are:

1) All of the above
2) 1, II and III
3) II, III and IV
4) Only I and III

52. Match the following columns and choose the correct option.

Column I		Column II	
1	Khasi and Jaintia Hills	a	Madhya Pradesh
2	Western ghat regions	b	Karnataka and Maharashtra
3	Sarguja, Chanda and Baster areas	c	Rajasthan
4	Aravalli hills	d	Meghalaya

1) 1-c, 2-b, 3a, 4-d
2) 1-a, 2-b, 3-d, 4-c

3) 1-d, 2-a, 3b, 4-c
4) 1-d, 2-b, 3a, 4-c

53. What % of the earth land area is covered by all biodiversity hotspot put together?
1) 6%
2) 10%
3) 15%
4) Less than 2%

54. Keystone species deserve protection because these
1) Are capable of surviving in harsh environmental conditions
2) Indicate presence of certain minerals in the soil
3) Have become rare due to overexploitation
4) Play an important role in supporting other species

55. Exploring molecular, genetic and species level diversity for product of economic importance is called
1) Agroforestry
2) Phytoremediation
3) Bioprospecting
4) Bioremediation

Answer Key

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
2	1	2	2	4	1	1	3	1	4	1	3	1	3	2
16	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
2	2	3	4	4	3	1	4	1	2	4	4	4	4	2
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**
4	3	3	4	2	4	4	3	1	1	1	4	4	1	4
46	**47**	**48**	**49**	**50**	**51**	**52**	**53**	**54**	**55**					
4	1	2	4	4	4	4	4	4	3					